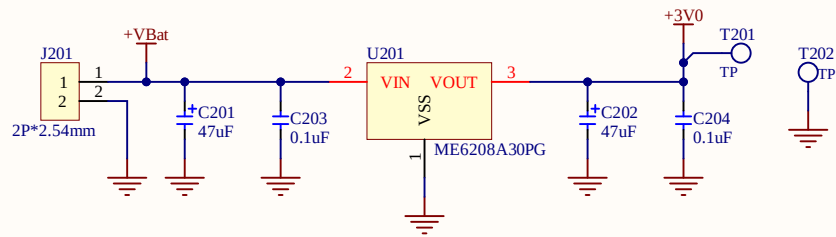


1 Power

Power.SchDoc



SmartWatch

Title: Smart Watch RS1.PrjPcb

Page Contents: Power.SchDoc

Size:

DWG NO

Revision:

*

Date: 5/10/2026

ThinhHC

Sheet 1 of

4

2 MCU

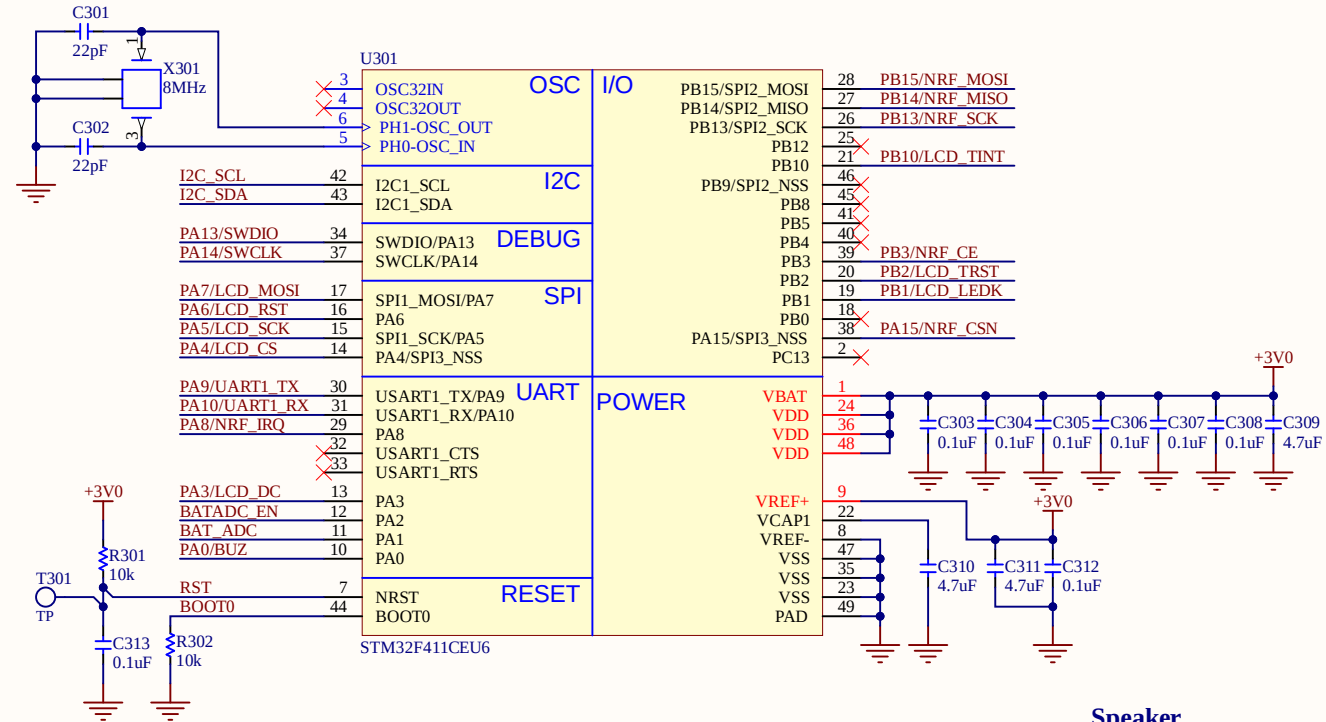
MCU.SchDoc

[2][3]NRF24L01

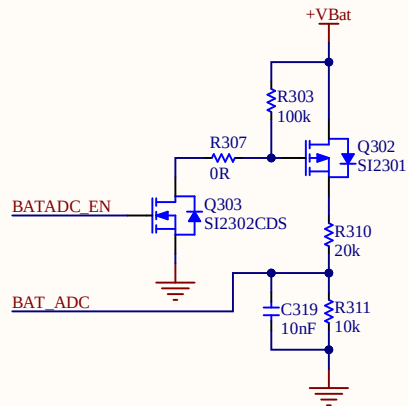
PB15/NRF_MOSI	PB15/NRF_MOSI
PB14/NRF_MISO	PB14/NRF_MISO
PB13/NRF_SCK	PB13/NRF_SCK
PB3/NRF_CE	PB3/NRF_CE
PA8/NRF_IRQ	PA8/NRF_IRQ
PA15/NRF_CSN	PA15/NRF_CSN

[2][4]TFT_LCD

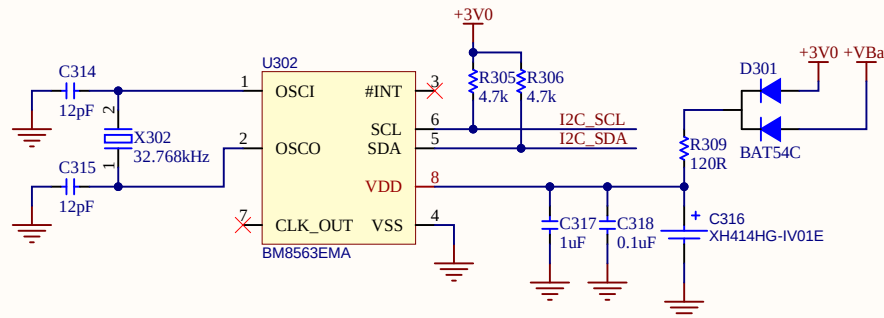
I2C_SCL	I2C_SCL
I2C_SDA	I2C_SDA
PA4/LCD_CS	PA4/LCD_CS
PA5/LCD_SCK	PA5/LCD_SCK
PA6/LCD_RST	PA6/LCD_RST
PA7/LCD_MOSI	PA7/LCD_MOSI
PB10/LCD_TINT	PB10/LCD_TINT
PB2/LCD_TRST	PB2/LCD_TRST
PA3/LCD_DC	PA3/LCD_DC
PB1/LCD_LEDK	PB1/LCD_LEDK



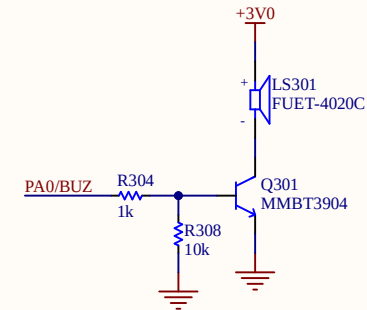
Battery monitoring



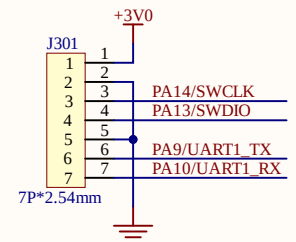
Logo QR



Speaker



Debug



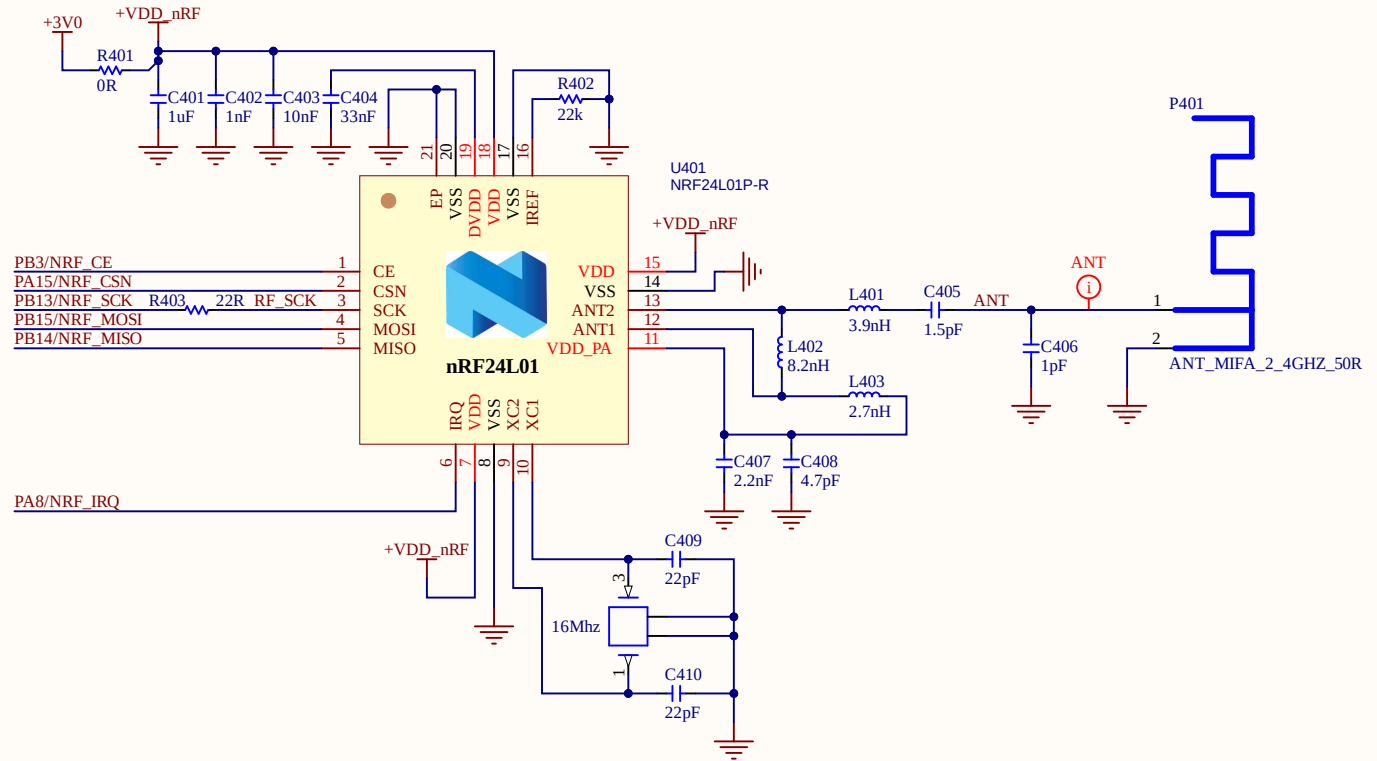
SmartWatch

Title: Smart Watch RS1.PrjPcb			
Page Contents: MCU.SchDoc			
Size:	DWG NO	Revision:	
Date: 5/10/2026	ThanhHC	Sheet 2	of 4

NRF24L01-M.SchDoc

[2][3]NRF24L01

PB15/NRF_MOSI	PB15/NRF_MOSI
PB14/NRF_MISO	PB14/NRF_MISO
PB13/NRF_SCK	PB13/NRF_SCK
PB3/NRF_CE	PB3/NRF_CE
PA8/NRF_IRQ	PA8/NRF_IRQ
PA15/NRF_CSN	PA15/NRF_CSN



SmartWatch

Title: Smart Watch RS1.PrjPcb

Page Contents: NRF24L01-M.SchDoc

Size:

DWG NO

Revision:

*

Date: 5/10/2026

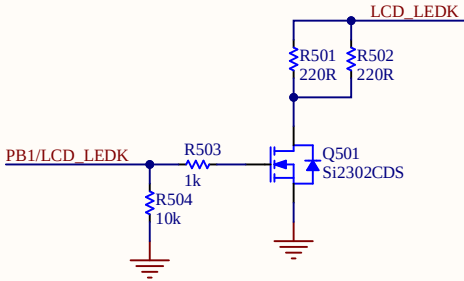
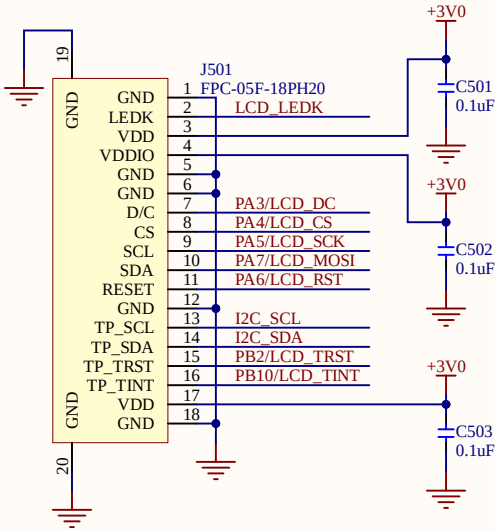
ThinhHC

Sheet 3 of 4

4 TFD_LCD

TFT_LCD.SchDoc

[2][4]TFT_LCD	
I2C_SCL	I2C_SCL
I2C_SDA	I2C_SDA
PA4/LCD_CS	PA4/LCD_CS
PA5/LCD_SCK	PA5/LCD_SCK
PA6/LCD_RST	PA6/LCD_RST
PA7/LCD_MOSI	PA7/LCD_MOSI
PB10/LCD_TINT	PB10/LCD_TINT
PB2/LCD_TRST	PB2/LCD_TRST
PA3/LCD_DC	PA3/LCD_DC
PB1/LCD_LEDK	PB1/LCD_LEDK



SmartWatch			
Title: Smart Watch RS1.PrjPcb			
Page Contents: TFT_LCD.SchDoc			
Size:	DWG NO		Revision: *
Date: 5/10/2026	ThinhHC		Sheet 4 of 4



RX

TX

GND

SWDIO

SWCLK

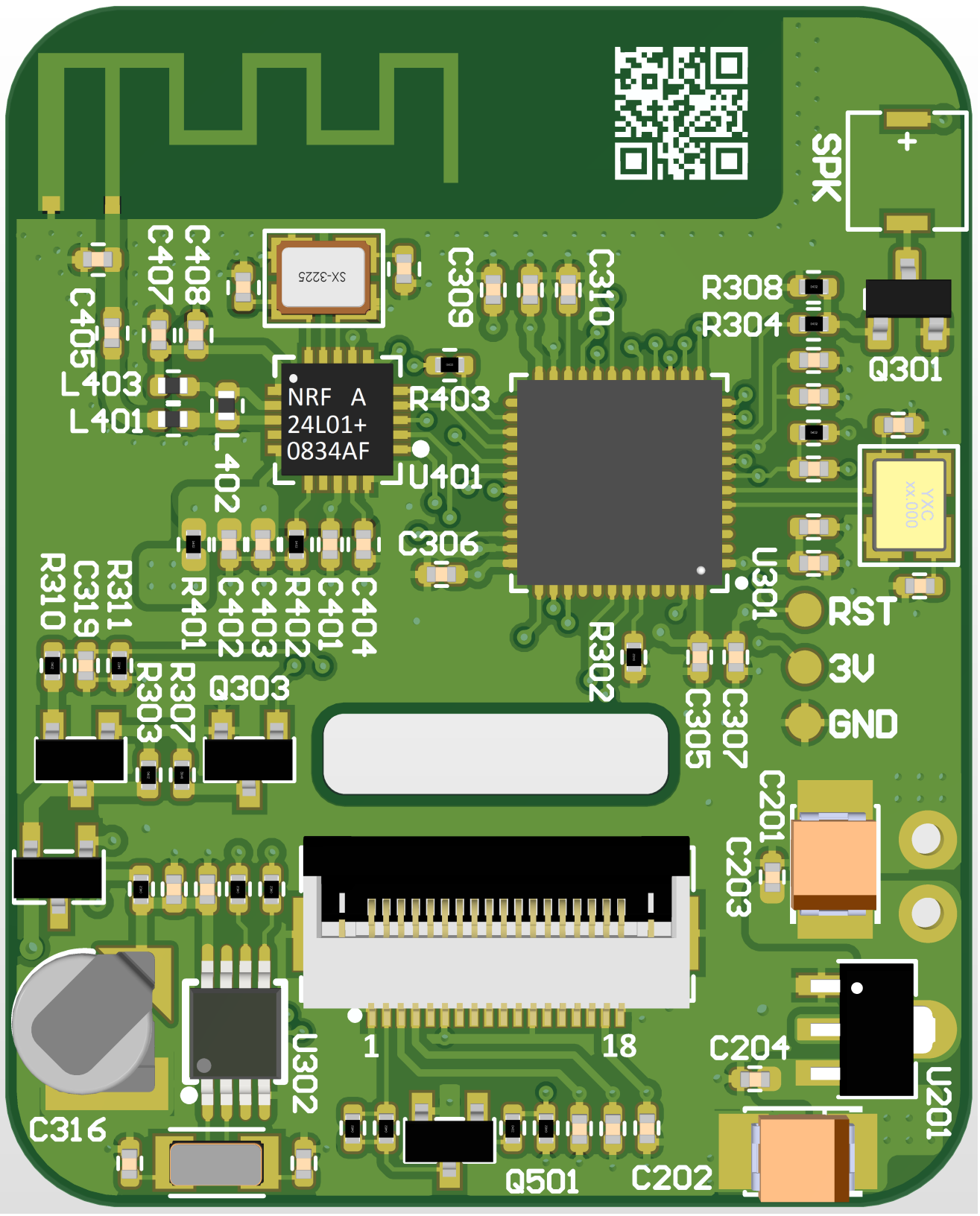
GND

3V

-VBAT

+VBAT





Smart Watch RS1.PrjPcb

Comment	Design	Footprint	IPC Part Num	Manufacturer Part Num	Manufacturer	Quantity	Description	Assembly Type	Link
1 47uF	C201, C	CASE-B_3528	C136658	CA45-B-10V-47uF-K	CEC(Shenzhen Zhenhua XinY	2	47uF 10V -55°C~+125°C	SMT Assembly	
2 0.1uF	C203, C	C_0402	C1525	CL05B104KO5NNNC	Samsung Electro-Mechanics	14	100nF 0402 16V X7R ±5%	SMT Assembly	
3 22pF	C301, C	C_0402	C1555	0402CG220J500NT	FH (Guangdong Fenghua Adv	4	22pF 0402 50V C0G ±5%	SMT Assembly	
4 4.7uF	C309, C	C_0402	C23733	CL05A475MP5NRNC	Samsung Electro-Mechanics	3	4.7uF 0402 10V X5R ±2%	SMT Assembly	
5 12pF	C314, C	C_0402	C1547	0402CG120J500NT	FH (Guangdong Fenghua Adv	2	12pF 0402 50V C0G ±5%	SMT Assembly	
6 1uF	C317, C	C_0402	C52923, C1573	CL05A105KA5NQNC, 0402	Samsung Electro-Mechanics,	2	1uF 0402 16V X7R ±10%	SMT Assembly	
7 10nF	C319, C	C_0402	C15195, C1524	CL05B103KB5NNNC, 0402	Samsung Electro-Mechanics,	2	10nF 0402 50V X7R ±1%	SMT Assembly	
8 1nF	C402	C_0402	C1523	0402B102K500NT	FH (Guangdong Fenghua Adv	1	1nF 0402 50V X7R ±10%	SMT Assembly	
9 33nF	C404	C_0402	C60425	0402B333K160NT	FH (Guangdong Fenghua Adv	1	33nF 0402 16V X7R ±1%	SMT Assembly	
10 1.5pF	C405	C_0402	C1552	0402CG1R5C500NT	FH (Guangdong Fenghua Adv	1	1.5pF 0402 50V C0G M	SMT Assembly	
11 1pF	C406	C_0402	C1550	0402CG1R0C500NT	FH (Guangdong Fenghua Adv	1	1pF 0402 50V C0G Mult	SMT Assembly	
12 2.2nF	C407	C_0402	C1531	0402B222K500NT	FH (Guangdong Fenghua Adv	1	2.2nF 0402 50V X7R ±1%	SMT Assembly	
13 4.7pF	C408	C_0402	C1569	0402CG4R7C500NT	FH (Guangdong Fenghua Adv	1	4.7pF 0402 50V C0G M	SMT Assembly	
14 BAT54C	D301	SOT-23-3_L2.9-W1.1	C14308	BAT54CLT1G	ON Semicon	1	SCHOTTKY BARRIER	SMT Assembly	
15 FPC-05F-18PH20	J501	FPC-SMD_18P-P0.5	C2856802	FPC-05F-18PH20	XUNPU	1	-25°C~+85°C 0.3mm 0.3	SMT Assembly	
16 3.9nH	L401	L_0402	C317939	FHW0402UC3N9JST	FH (Guangdong Fenghua Adv	1	3.9nH 840mA ±5% 66m	SMT Assembly	
17 8.2nH	L402	L_0402	C317946	FHW0402UC8N2JST	FH (Guangdong Fenghua Adv	1	8.2nH 680mA ±5% 100	SMT Assembly	
18 2.7nH	L403	L_0402	C317935	FHW0402UC2N7KST	FH (Guangdong Fenghua Adv	1	2.7nH 640mA ±10% 12	SMT Assembly	
19 FUET-4020C	LS301	Buzzer SMD	C41410568	FUET-4020C	FUET	1	-30°C~+70°C 100mA 2V	SMT Assembly	
20 MMBT3904	Q301	SOT-23-3	C20526	MMBT3904	Changjiang Electronics Tech (	1	MMBT3904 TRANSISTO	SMT Assembly	
21 SI2301	Q302	SOT-23-3_L2.9-W1.1	C10487	SI2301CDS-T1-GE3	Vishay Intertech	1	1 P-Channel 1.6W -55°	SMT Assembly	
22 SI2302CDS	Q303, Q	SOT-23-3_L2.9-W1.1	C10488	SI2302CDS-T1-GE3	Vishay Intertech	2	1 N-channel 2.9A 20V 3	SMT Assembly	
23 10k	R301, R	R_0402	C25744	0402WGF1002TCE	UNI-ROYAL(Uniroyal Elec)	5	10kΩ 0402 50V 62.5mW	SMT Assembly	
24 100k	R303	R_0402	C25741	0402WGF1003TCE	UNI-ROYAL(Uniroyal Elec)	1	100kΩ 0402 50V -55°C~	SMT Assembly	
25 1k	R304, R	R_0402	C11702	0402WGF1001TCE	UNI-ROYAL(Uniroyal Elec)	2	1kΩ 0402 50V 62.5mW	SMT Assembly	
26 4.7k	R305, R	R_0402	C25900	0402WGF4701TCE	UNI-ROYAL(Uniroyal Elec)	2	4.7kΩ 0402 50V -55°C~	SMT Assembly	
27 0R	R307, R	R_0402	C17168	0402WGF0000TCE	UNI-ROYAL(Uniroyal Elec), Un	2	0Ω 0402 50V -55°C~+1	SMT Assembly	
28 120R	R309	R_0402	C25079	0402WGF1200TCE	UNI-ROYAL(Uniroyal Elec)	1	120Ω 0402 50V -55°C~	SMT Assembly	
29 20k	R310	R_0402	C25765	0402WGF2002TCE	UNI-ROYAL(Uniroyal Elec)	1	20kΩ 0402 50V 62.5mW	SMT Assembly	
30 22k	R402	R_0402	C25768	0402WGF2202TCE	Uniroyal Elec	1	22kΩ 0402 62.5mW Th	SMT Assembly	
31 22R	R403	R_0402	C25092	0402WGF220JTCE	Uniroyal Elec	1	22Ω 0402 62.5mW Thic	SMT Assembly	
32 220R	R501, R	R_0402	C25091	0402WGF2200TCE	UNI-ROYAL(Uniroyal Elec)	2	220Ω 0402 50V -55°C~	SMT Assembly	
33 ME6208A30PG	U201	SOT_89	C80196	ME6208A30PG	MICRONE(Nanjing Micro One	1	-45°C~+150°C 1 150mA	SMT Assembly	
34 STM32F411CEU6	U301	UFQFPN-48_L7.0-W	C60420	STM32F411CEU6	STMicroelectronics	1	-40°C~+85°C 1.7V~3.6V	SMT Assembly	
35 BM8563EMA	U302	MSOP-8_L3.0-W3.0	C269878	BM8563EMA	GATEMODE(捷茂微)	1	MSOP-8 Real-time Clo	SMT Assembly	
36 NRF24L01P-R	U401	QFN-20_L4.0-W4.0	C8791	NRF24L01P-R	NORDIC	1	NRF24L01P-R SPI QFN	SMT Assembly	
37 8MHz	X301	CRYSTAL_SMD3225	C2682774	X32258MSB4SI	YXC Crystal Oscillators	1	8MHz 20pF 120Ω -40°C	SMT Assembly	
38 32.768kHz	X302	CRYSTAL-SMD_L3.2	C841881	Q13FC13500005	Seiko Epson	1	32.768kHz 12.5pF ±10p	SMT Assembly	
39 16Mhz	X401	SMD-3225_4P	C13738	X322516MLB4SI	Yangxing Tech	1	16MHz SMD Crystal Re	SMT Assembly	